Calculus II

Problems for Training #6 24–28 Mar 2025

Topics:

- Evaluating iterated integrals
- Evaluating double integrals over rectangles
- Evaluating double integrals over non-rectangles
- Area by double integrals

Problem 1. Evaluate the iterated integral.

$$\int_{-1}^{2}\int_{1}^{2} x\ln y\,dy\,dx.$$

Problem 2. Evaluate the iterated integral.

$$\iint_R xy\cos y\,dA, \quad R: \quad -1\le x\le 1, \quad 0\le y\le \pi.$$

In Problems 3–4, integrate f over the given region.

Problem 3. Find the volume of the region bounded above by the paraboloid $z = x^2+y^2$ and below by the square $R: -1\le x\le 1,\ -1\le y\le 1$.

Problem 4. Find the volume of the region bounded above by the plane $z = 2-x-y$ and below by the square $R:\ 0\le x\le 1,\ 0\le y\le 1$.

Problem 5. *Minimizing a double integral.* What region R in the $xy-$plane minimizes the value of

$$\iint_R (x^2+y^2-9)\,dA?$$

Justify your answer.

Problem 6. Sketch the region bounded by the given lines and curves. Then express the region's area as an iterated double integral and evaluate the integral.

The lines $y = 2x$, $y = x/2$, and $y = 3-x$.

Problem 7. The integrals and sums of integrals in the problem give the areas of regions in the $xy-$plane. Sketch each region, label each bounding curve with its equation, and

give the coordinates of the points where the curves intersect. Then find the area of the region.

$$\int_{-1}^{0}\int_{-2x}^{1-x} dy\,dx + \int_{0}^{2}\int_{-x/2}^{1-x} dy\,dx.$$

Problem 8. Find the average value of $f(x,y) = 1/(xy)$ over the square $\ln 2 \le x \le 2\ln 2$, $\ln 2 \le y \le 2\ln 2$.

END

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Answers to selected problems

1. $\frac{3(2\ln 2-1)}{2}$
2. 0
3. $\frac{8}{3}$
4. 1
6. $\frac{3}{2}$
7. $\frac{3}{2}$
8. 1